

CMTA Project No.:	B069-02-039	Control No.:	H-02.00
CMTA Project Name:	Boston Properties @ 7 Times Square - Snowflake Floors 29-31	Page:	1 of 1
☐ NO EXCEPTIONS TAKEN No further review of submittal required.		☐ REVISE AND RESUBMIT Revise as noted and resubmit for further review.	
☒ MAKE CORRECTIONS AS NOTED Incorporate corrections in work; resubmission is not required. Where corrections cannot be complied with, revise the exceptions and resubmit.		☐ EXAMINED Submittal received, formal review by others and/or formal review by this office not required. Submittal has been examined for information only.	
Corrections or comments made on the shop drawings during this review do not relieve contractor from compliance with requirements of the drawings and specifications. This check is only for review of general conformance with the design concept of the project and general compliance with the information given in the contract documents. The contractor is responsible for: confirming and correlating all quantities and dimensions; selecting fabrication processes and techniques of construction; coordinating his work with that of all other trades; and performing his work in a safe and satisfactory manner.			
Date:	6/15/2026	Engineer:	Mike Sisti
SUBMITTAL DESCRIPTION:	Pump Cuts		
COMMENTS: CMTA Comments: 1. Provide vibration isolation for pumps and piping within 50 feet or 50 pipe diameters, whichever is greater. See specifications for details. 2. Provide valve and trim as per the details on the design documents. 3. Pumps shall meet all pressure and flow characteristics as per the design documents. 4. Contractor to provide the differential pressure at the tie in to the risers in full flow condition, as per the design documents, and provide to the manufacturer to ensure the proper head is obtained to ensure full flow throughout the system.			



Northern Wolves AC
55 9th St, 55-A2
Brooklyn, New York 11215
P: (347) 463-9248

Project: 11010174 Snowflake Fitout - 7 Times Square 29-31st Floor
7 Times Square
New York, New York 10036

Submittal #2.0 - PUMPS - Grundfos

Revision	0	Submittal Manager	Kastriot Cikaqi (Northern Wolves, Inc.)
Status	Open	Date Created	Jun 9, 2026
Issue Date	Jun 9, 2026	Spec Section	
Responsible Contractor		Received From	
Received Date		Submit By	
Final Due Date		Lead Time	42 day(s)
		Cost Code	
Location		Type	
Submittal Package			
Approvers			
Ball in Court	Kastriot Cikaqi (Northern Wolves, Inc.)		
Distribution			
Description			

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					Submittal - Snowflake.pdf

Dolphin Equipment Corp.
Tel# 914-663-8355**GRUNDFOS**

Submittal Data

PROJECT: Snowflake 7 Times Sq [fl. 29-31]	UNIT TAG: CWP-29-1,2; 30-1,2; 31-1,2	QUANTITY: 6
REPRESENTATIVE: Dolphin Equipment	TYPE OF SERVICE:	
ENGINEER: CMTA	SUBMITTED BY: Barry Brar	DATE: 6/9/26
CONTRACTOR: Northern Wolves AC	APPROVED BY:	DATE:
	ORDER NO.:	DATE:

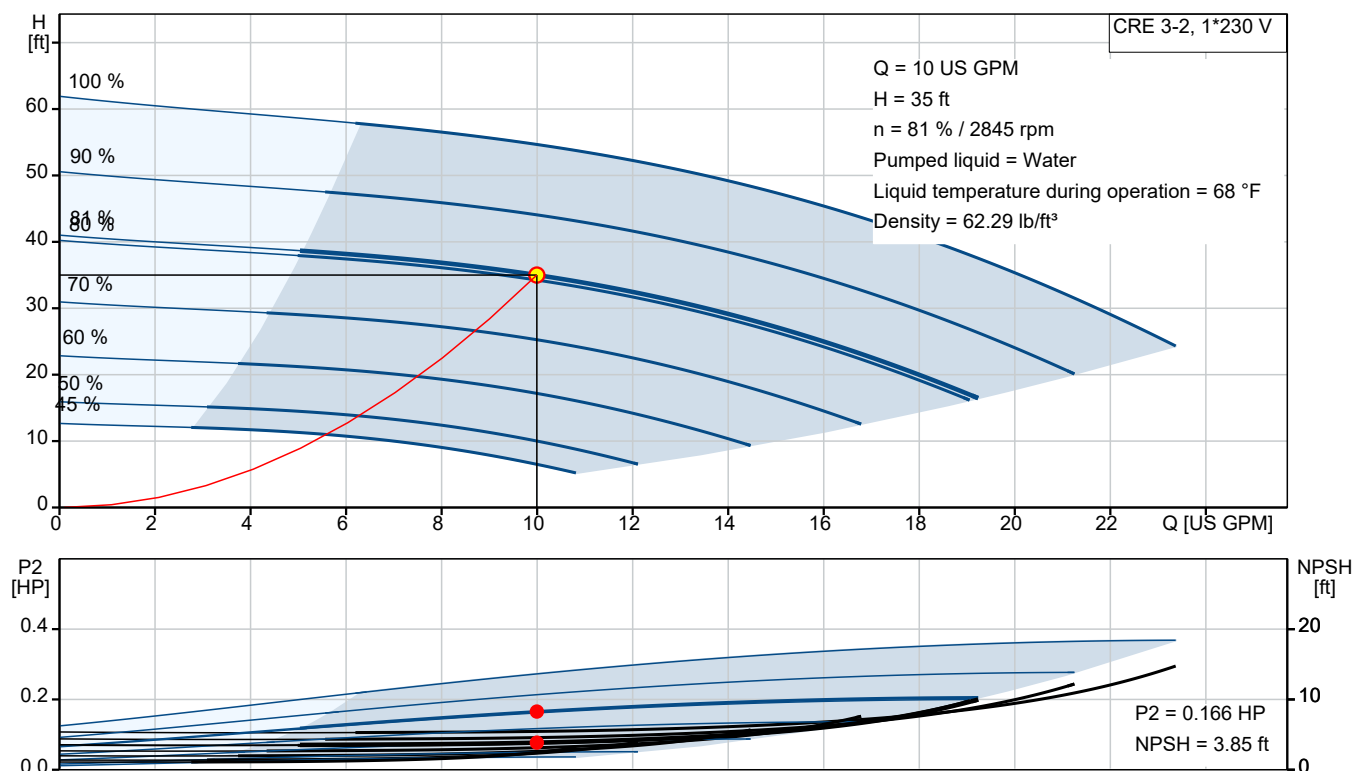


Vertical, multistage centrifugal pump with integrated frequency converter. The pump head and base are in cast iron - all other wetted parts are in stainless steel (EN 1.4301)

Pump Model CRE3-DOL2VFD with Integrated VFD,
10gpm, 35ft tdh, 0.5hp, 2845rpm, 208v/1ph Motor

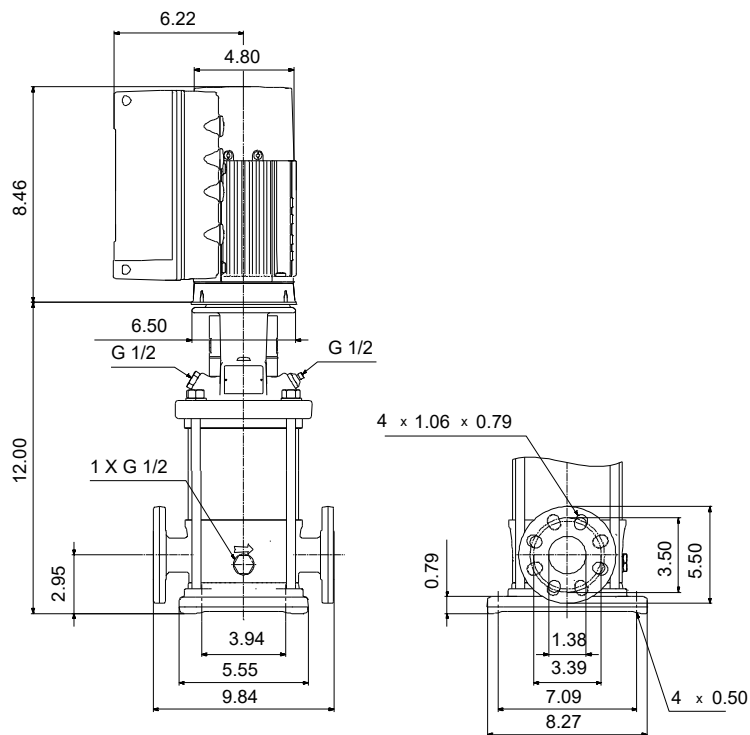
Note! Product picture may differ from actual product

Conditions of Service	Pump Data	Motor Data
Flow: 10 US GPM	Max pressure at stated temp: 363 psi / 250 °F	Rated power - P2: 0.5 HP
Head: 35 ft	Liquid temperature range: -4 .. 248 °F	Rated voltage: 200-240 V
Efficiency: 38.5 %	Maximum ambient temperature: 122 °F	Mains frequency: 60 Hz
Liquid: Water	Shaft seal: HQQE	Enclosure class: IP55
Temperature: 68 °F	Product number: On request	Insulation class: F
NPSH required: 3.854 ft		Motor protection: ELEC
Specific Gravity: 1.000		Motor type: 71A
		Eta 1/1: 84 %





Submittal Data

**Materials:**

Base:	Cast iron EN 1561 EN-GJL-200 ASTM A48-25B
Impeller:	Stainless steel AISI 304 EN 1.4301
Material code:	A
Code for rubber:	E



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Phone:

Date:

03/04/2026

Qty.	Description
1	CRE 3-2 A-FGJ-A-E-HQQE  Note! Product picture may differ from actual product Product No.: On request Vertical, multistage centrifugal pump with inlet and outlet ports on same the level (inline). The pump head and base are in cast iron – all other wetted parts are in stainless steel. A cartridge shaft seal ensures high reliability, safe handling, and easy access and service. Power transmission is via a rigid split coupling. Pipe connection is via combined DIN-ANSI-JIS flanges. The pump is fitted with a 1-phase, fan-cooled, permanent-magnet, synchronous motor. The motor efficiency is classified as IE5 in accordance with IEC 60034-30-2. The motor includes a frequency converter and PI controller in the motor terminal box. This enables continuously variable control of the motor speed, which again enables adaptation of the performance to a given requirement. The operating panel on the motor terminal box features a four-inch TFT display, push-buttons and the Grundfos Eye indicator. The display gives an intuitive and user-friendly interface to all functions. The push-buttons are used to navigate through the menu structure to access pump and performance data on site and enable setting of required setpoint as well as setting of pump to "Min." or "Max." operation or to "Stop". The Grundfos Eye indicator on the operating panel provides visual indication of pump status: • "Power on": Motor is running (rotating green indicator lights) or not running (permanently green indicator lights) • "Warning": Motor is still running (rotating yellow indicator lights) or has stopped (permanently yellow indicator lights) • "Alarm": Motor has stopped (flashing red indicator lights). Communication with the pump is also possible by means of Grundfos GO Remote (accessory). The remote control enables further settings as well as reading out of a number of parameters such as "Actual value", "Speed", "Power input" and total "Power consumption". The terminal box has a number of inputs and outputs enabling the motor to be used in advanced applications where many inputs and outputs are required: • two dedicated digital inputs • three analog inputs, 0(4)-20 mA, 0-5 V, 0-10 V, 0.5-3.5 V • 5 V voltage supply to potentiometer and sensor • one analog output, 0-10 V, 0(4)-20 mA • two configurable digital inputs or open-collector outputs • two Pt100/Pt1000 inputs • LiqTec, dry-running protection sensor input • Grundfos Digital Sensor input and output • 24 V voltage supply for sensors • two signal-relay outputs (potential-free contacts) • GENIbus connection • interface for Grundfos CIM fieldbus module. Further product details An external sensor can be connected if controlled pump operation based on for example flow, differential pressure or temperature is required. The operating panel on the motor terminal box features a four-inch TFT display, push-buttons and the Grundfos Eye indicator.



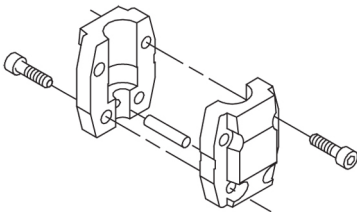
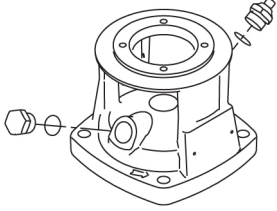
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Qty.	Description
1	The display gives an intuitive and user-friendly interface to all functions. The push-buttons are used to navigate through the menu structure to access pump and performance data on site and enable setting of required setpoint as well as setting of pump to "Min." or "Max." operation or to "Stop". The Grundfos Eye indicator on the operating panel provides visual indication of pump status: • "Power on": Motor is running (rotating green indicator lights) or not running (permanently green indicator lights) • "Warning": Motor is still running (rotating yellow indicator lights) or has stopped (permanently yellow indicator lights) • "Alarm": Motor has stopped (flashing red indicator lights). Communication with the pump is also possible by means of Grundfos GO Remote (accessory). The remote control enables further settings as well as reading out of a number of parameters such as "Actual value", "Speed", "Power input" and total "Power consumption". Steel, cast iron and aluminium components have an epoxy-based coating made in a cathodic electro-deposition (CED) process. CED is a high-quality dip-painting process where an electrical field around the products ensures deposition of paint particles as a thin, well-controlled layer on the surface. An integral part of the process is a pretreatment. The entire process consists of these elements: 1) Alkaline-based cleaning. 2) Zinc phosphating. 3) Cathodic electro-deposition. 4) Curing to a dry film thickness 18-22 my m. The colour code for the finished product is NCS 9000/RAL 9005. Pump A standard split coupling connects the pump and motor shaft. It is enclosed in the pump head/motor stool by means of two coupling guards.  The pump head, pump head cover and flange for motor mounting is made in one piece. The pump head has a combined 1/2" priming plug and vent screw.  The pump is fitted with a balanced O-ring seal unit with a rigid torque-transmission system. This seal type is assembled in a cartridge unit which makes replacement safe and easy. Due to the balancing, this seal type is suitable for high-pressure applications. The cartridge construction also protects the pump shaft from possible wear from a dynamic O-ring between pump shaft and shaft seal. Seal faces: • Rotating seal ring material: silicon carbide (SiC) • Stationary seat material: silicon carbide (SiC) This material pairing is used where higher corrosion resistance is required. The high hardness of this material pairing offers good resistance against abrasive particles. Secondary seal material: EPDM (ethylene-propylene rubber) EPDM has excellent resistance to hot water. EPDM is not suitable for mineral oils.



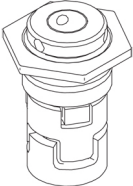
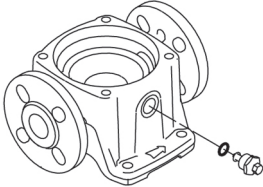
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Qty.	Description
1	 The shaft seal is screwed into the pump head. The chambers and impellers are made of stainless-steel sheet. The chambers are provided with a PTFE neck ring offering improved sealing and high efficiency. The impellers have smooth surfaces, and the shape of the blades ensure a high efficiency. The base is made of cast iron. The flanges and base are cast in one piece. The outlet side of the base has a combined drain plug and bypass valve. The pump is secured to the foundation by four bolts through the base plate.  Motor The motor is a totally enclosed, fan-cooled motor with principal dimensions to IEC and DIN standards. The motor is flange-mounted with tapped-hole flange (FT). Motor-mounting designation in accordance with IEC 60034-7: IM B 14 (Code I) / IM 3601 (Code II). Electrical tolerances comply with IEC 60034. The motor efficiency is classified as IE5 in accordance with IEC 60034-30-2. The motor requires no external motor protection. The motor control unit incorporates protection against slow- and quick-rising temperatures, e.g. constant overload and stalled conditions. The terminal box has a number of inputs and outputs enabling the motor to be used in advanced applications where many inputs and outputs are required: • two dedicated digital inputs • three analog inputs, 0(4)-20 mA, 0-5 V, 0-10 V, 0.5-3.5 V • 5 V voltage supply to potentiometer and sensor • one analog output, 0-10 V, 0(4)-20 mA • two configurable digital inputs or open-collector outputs • two Pt100/Pt1000 inputs • LiqTec, dry-running protection sensor input • Grundfos Digital Sensor input and output • 24 V voltage supply for sensors • two signal-relay outputs (potential-free contacts) • GENIbus connection • interface for Grundfos CIM fieldbus module. Technical data Liquid: Pumped liquid: Water Liquid temperature range: -4 .. 248 °F Selected liquid temperature: 68 °F Density: 62.29 lb/ft³



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Qty.	Description
1	Technical: Pump speed on which pump data are based: 2845 rpm Actual calculated flow: 10 US GPM Resulting head of the pump: 35 ft Pump orientation: Vertical Shaft seal arrangement: Single Primary shaft seal: HQQE Code for shaft seal: HQQE Approvals: CE,UKCA,SEPRO,RCM, Approvals for drinking water: NSF/ANSI 61 Curve tolerance: ISO9906:2012 3B Materials: Type key, code for materials: A Type key, code for rubber components. E = EPDM, V=FKM: E Base: Cast iron EN 1561 EN-GJL-200 ASTM A48-25B Impeller: Stainless steel EN 1.4301 AISI 304 Bearing: SIC Installation: Maximum ambient temperature: 122 °F Maximum operating pressure: 362.59 psi Max pressure at stated temp: 363 psi / 250 °F 363 psi / -4 °F Type of connection: DIN / ANSI / JIS Size of inlet connection: DN 25/32 1 1/4 inch Size of outlet connection: DN 25/32 1 1/4 inch Pressure rating for connection: PN 25 Flange rating inlet: 250 lb Flange size for motor: 56C Terminal box position: 6 Electrical data: Motor standard: NEMA Motor type: 71A Rated power - P2: 0.5 HP Power (P2) required by pump: 0.5 HP Over/undersize motor: Standard motor size Mains frequency: 60 Hz Rated voltage: 1 x 200-240 V Service factor: 1.00 Rated current: 2.4-2.1 A



Company name:

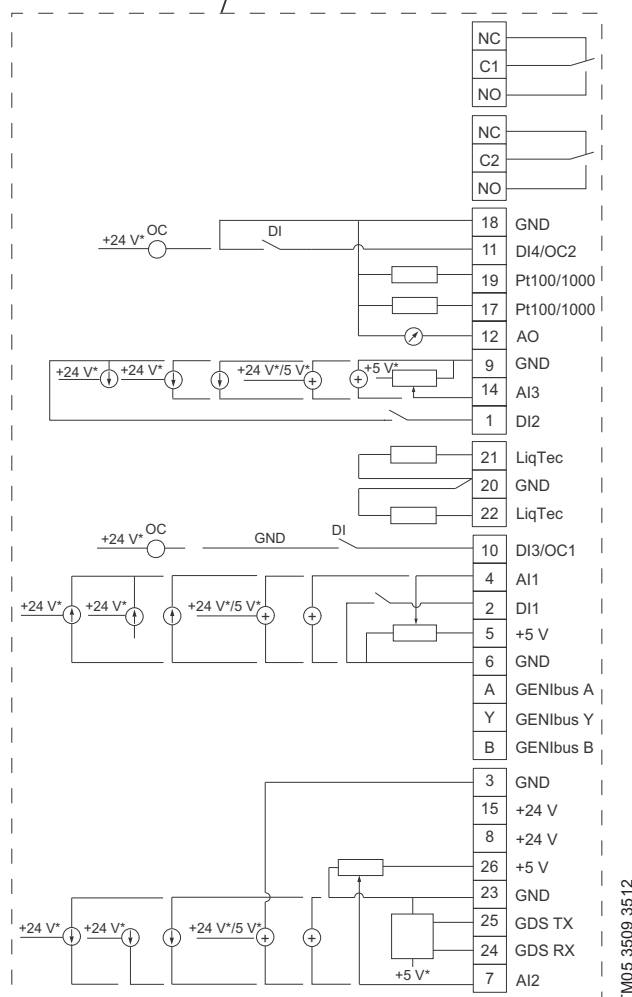
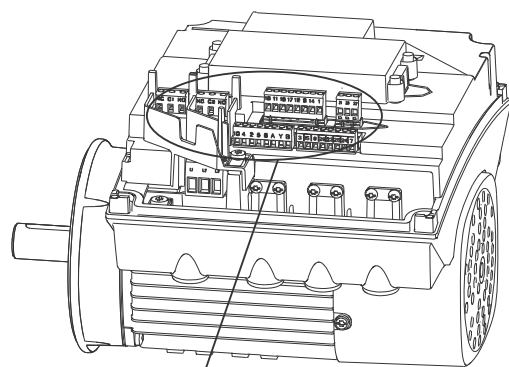
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Qty.	Description
1	Cos phi - power factor:0.96 Rated speed:360-4000 rpm IE Efficiency class:IE5 Motor efficiency at full load:84 % Enclosure class (IEC 34-5):IP55 Insulation class (IEC 85):F Motor No:98362269 Controls: Terminal box position:6 Frequency converter:Built-in Pressure sensor:N Others: Net weight:61.8 lb Gross weight:72.8 lb Shipping volume:4.94 ft³ Country of origin:US Custom tariff no.:8413.70.2040



TM05 3509 3512

* If you use an external supply source, there must be a connection to GND.

Fig. 8 Connection terminals, CRE, CRIE, CRNE, CRKE, SPKE, MTRE and CME pumps

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1 (LIVE or PELV)
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2 (PELV only)
C2	Common	
NO	Normally open contact	
18	GND	Ground
11	DI4/OC2	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
19	Pt100/1000 input 2	Pt100/1000 sensor input
17	Pt100/1000 input 1	Pt100/1000 sensor input
12	AO	Analog output: 0-20 mA / 4-20 mA 0-10 V
9	GND	Ground
14	AI3	Analog input: 0-20 mA / 4-20 mA 0-10 V
1	DI2	Digital input, configurable
21	LiqTec sensor input 1	LiqTec sensor input (white conductor)
20	GND	Ground (brown and black conductors)
22	LiqTec sensor input 2	LiqTec sensor input (blue conductor)
10	DI3/OC1	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
4	AI1	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V
2	DI1	Digital input, configurable
5	+5 V	Supply to potentiometer and sensor
6	GND	Ground
A	GENIbus, A	GENIbus, A (+)
Y	GENIbus, Y	GENIbus, GND
B	GENIbus, B	GENIbus, B (-)
3	GND	Ground
15	+24 V	Supply
8	+24 V	Supply
26	+5 V	Supply to potentiometer and sensor
23	GND	Ground
25	GDS TX	Grundfos Digital Sensor output
24	GDS RX	Grundfos Digital Sensor input
7	AI2	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V